Analysis and Documentation:

This project performs Analysis using Python (Pandas, Matplotlib, Seaborn) and SQL (`pandasql`) on the Telco Customer Churn dataset.

Dataset Overview & Data Cleaning Workflow:

The dataset consists of **7,043 rows** (individual customers) and **21 features**

Telco Customer Churn

Dataset Structure & Schema:

- Customers who left : the column is called Churn
- Services that each customer has signed up for : phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies
- Account Info:enure, Contract, PaperlessBilling, PaymentMethod
- Services: PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection ,TechSupport, StreamingTV, StreamingMovies
- Financials :MonthlyCharges, TotalCharges
- Target Variable: churn (yes/no)

Data Cleaning & Transformation Steps

A. Handling Whitespaces and Data Types

As whitespaces are not considered empty values i checked for them first and found 11 whitespaces in the TotalCharges column

During the first inspection (df.info()), the TotalCharges column was found to have an object data type instead of a numeric type

Action Taken:

Removed extra spaces from the values
Converted the column to numeric using pd.to_numeric()

B. Imputation of Missing Values:

The 11 NaN values in TotalCharges belonged exclusively to new customers with a tenure of 0 months.

- Logic: Customers with 0 months of tenure have not yet accumulated billings
- Action Taken: Filled all 11 NaN values with 0.0.

C. Outlier Investigation (Modified Z-Score)

Applying the Modified Z-Score method shows 136 potential outliers in TotalCharges.

- From the business logic it make sense that these are customers that have high tenure may have high totalcharges because of the long time of subscribtions
- In the notebook visulization the red dots (outliers) are all clustered way up at the very top of the TotalCharges, and it clearly shows that they aren't random mistakes;
  These are the people with the highest Value for the company

If outlier handling is necessary, regardless outliers are valid according to business logic, I would apply log transformation this will help me compress the scale of TotalCharges, reducing the relative differences between large and small values while preserving all customer records , making extreme values have less impact for ML algorithms that will be used later

I would use capping or upper and lower bound it would change the numbers totally which i not support

D. Churn Imbalance Analysis:

- (No): 5,174 customers (73.46%)
- (Yes): 1,869 customers (26.54%)
- Result interpreting : The target variable demonstrates a class imbalance
- For further , If training a model later the data would better be balanced (will be applied in the machine learning notebook)

Examples of some columns impact :

Impact of Internet Service on Churn :

- DSL Churn Volume: 459 customers
- Fiber Optic Churn Volume: 1,297 customers
- No Internet Service Churn Volume: 113 customers

Insight: Fiber optic users are churning at a rate much higher compared to DSL users

The count /number of customers in each contract :

- 31.04% of customers have been with the company for 12 months or less
- 21.06% have maintained their subscriptions for at least 60 months.
- The result knowing the count of customer in each type of contract

Monthly Billing Impact (boxplot) :

- Customers who churned had higher monthly charges than those who stayed
- The average monthly charge was 74.4 for churned customers and 61.2 for non-churned
- customers who canceled their subscription have a higher average monthly bill (median around 80)
- This means that higher monthly costs are one of the main drivers of churn

SQL Data Analysis & Relational Joins (`pandasql`)

To simulate enterprise database environments, relational SQL queries were run directly on the DataFrames using `pandasql`.

Query 1: Customer distribution by contract type

Goal: subscribers grouped by their contract model

Churn across internet service types

Goal: Determine churn counts across different internet services types

Query 3: Multi-table relational LEFT JOIN

Goal: To practice I splitted the data to 2 tables

goal to see combination between the contract and internet service that a certain customers have and are these who churn

For further sight when training ML algorithm i may need to apply one-hot encoding (week3)